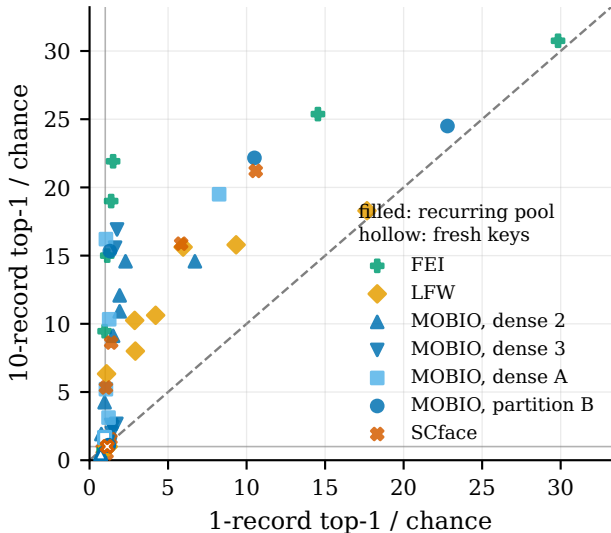
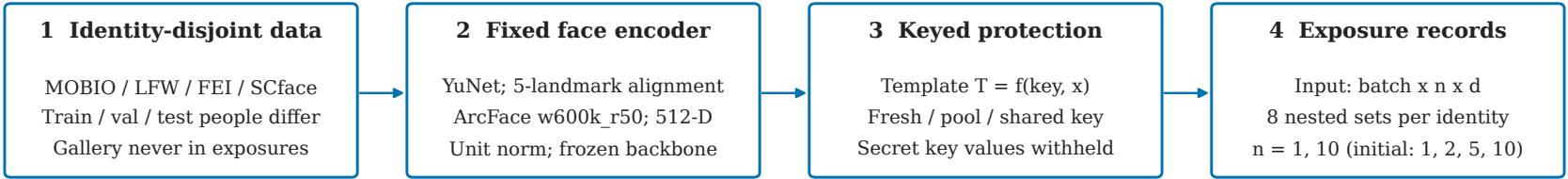


BioHash record-count gains



A DATA AND PROTECTION



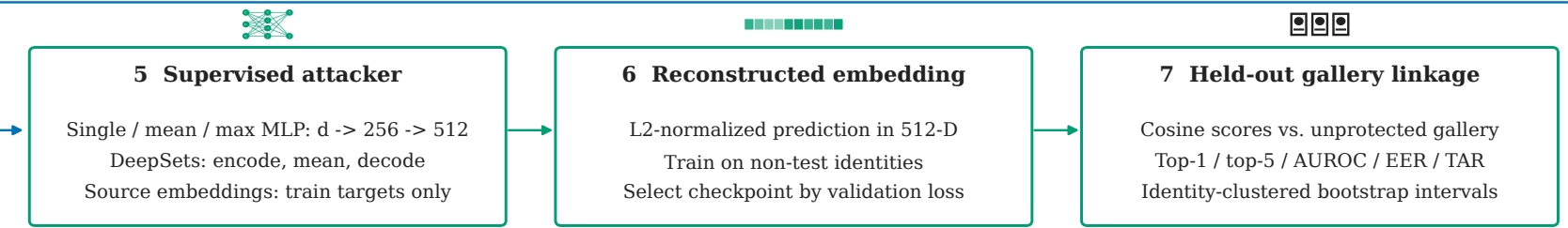
OUTPUT d

BioHash: 128 bits | MLP-Hash: 512 bits | IoM-GRP: 300 codes ($q=16$; one-hot $d=4,800$)

PolyProtect: 170 reals (window $m=5$, $\text{overlap}=2$). Scheme coverage differs by dataset.

B KEY-BLIND ATTACK AND EVALUATION

Attacker observes protected records, not keys or slot labels; slots are revealed only in a labelled control.



KEY DESIGN

Fresh: source-record keys are split-disjoint. Pool K: hidden transforms recur across identity splits.

REPLICATION

MOBIO / FEI follow-up: 2 identity partitions x 3 model seeds; matched 120-epoch training caps.

DESIGN

Same-identity aggregation under controlled transform reuse; detailed attacker paths in the companion figure.

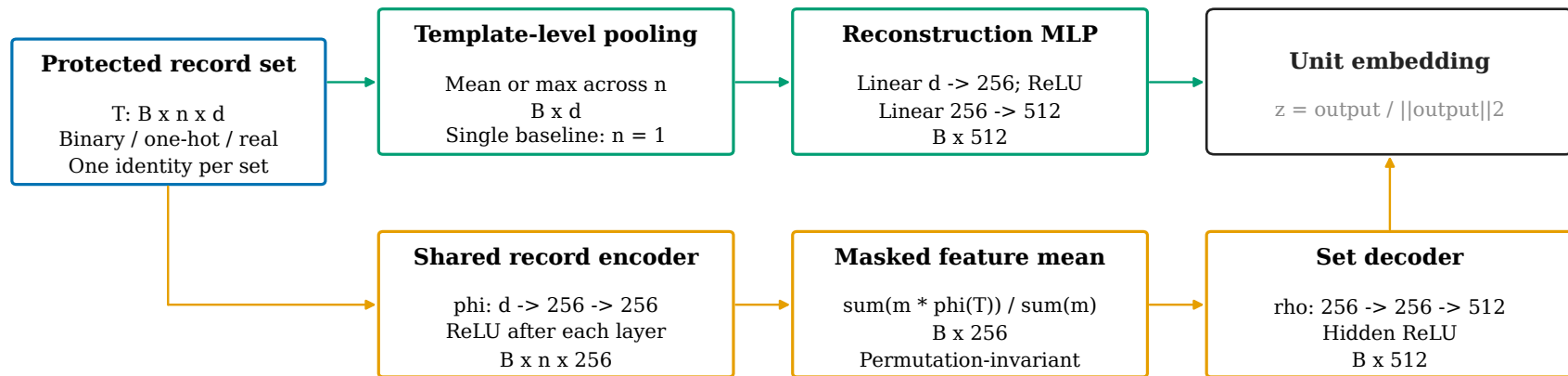
A MATCHED EXPOSURE CONSTRUCTION

One identity, n source records; nested subsets from one fixed permutation per repeat; gallery excluded.

Fresh: independent source-record keys. Pool K: hidden transforms recur across identity splits. No keys or slots in model input.

T: $B \times n \times d$ $d = 128$ BioHash; 512 MLP-Hash; 4,800 one-hot IoM-GRP; 170 PolyProtect

B TWO AGGREGATION PATHS



Primary: mean-pool MLP. Secondary: DeepSets. m masks valid records; all n records are valid in these runs.

C SUPERVISION

Target $u = \text{normalize}(\text{mean of exposed source embeddings})$.

Loss = $\text{mean}[1 - \cos(\text{z}, u)] + 0.1 \times \text{MSE}(\text{z}, u)$.

Adam: lr 0.001; weight decay 0.0001; hidden width 256.

Train identities only; select minimum validation loss; freeze test.

D HELD-OUT IDENTITY LINKAGE

Scores = $z G^T$; $G: N \times 512$ unprotected unit gallery.

Rank identities; top-1/top-5, AUROC, EER, TAR at FAR.

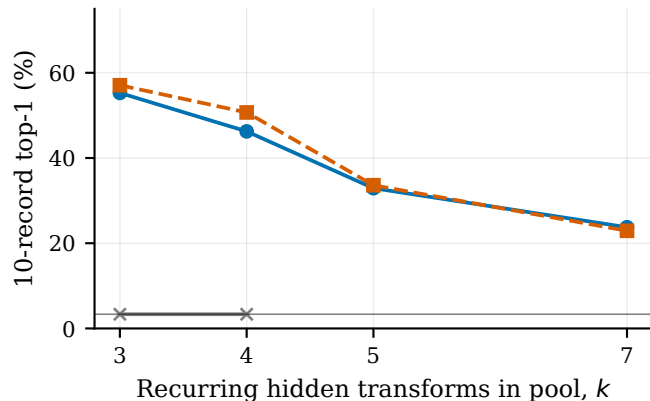
Follow-up: 3 model seeds x 2 identity partitions; 120 epochs.

Crossed seed/identity intervals; paired primary contrasts.

B: batch size; n: exposure count; d: template input width; N: test gallery identities. Source targets are not inference inputs.

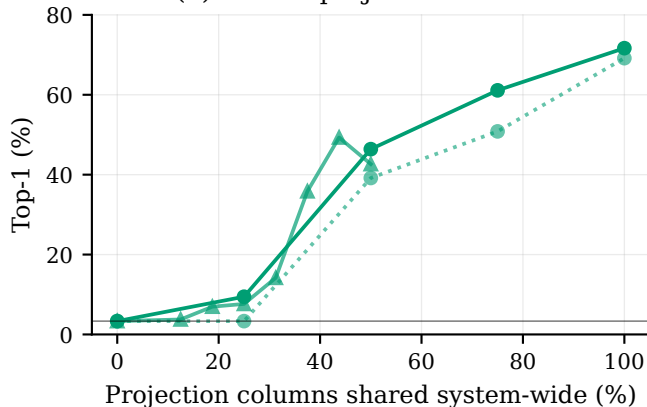
MOBIO mechanism controls (30 gallery identities; chance 3.33%)

(a) Slot label vs. shuffled records



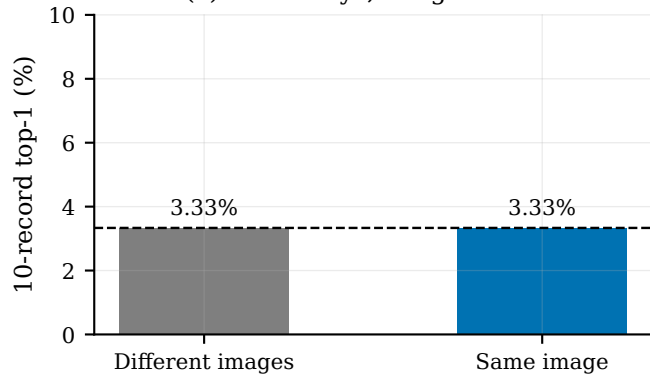
- slot hidden (DeepSets)
- slot label given to attacker
- records 2-10 from other identities

(b) Shared projection columns

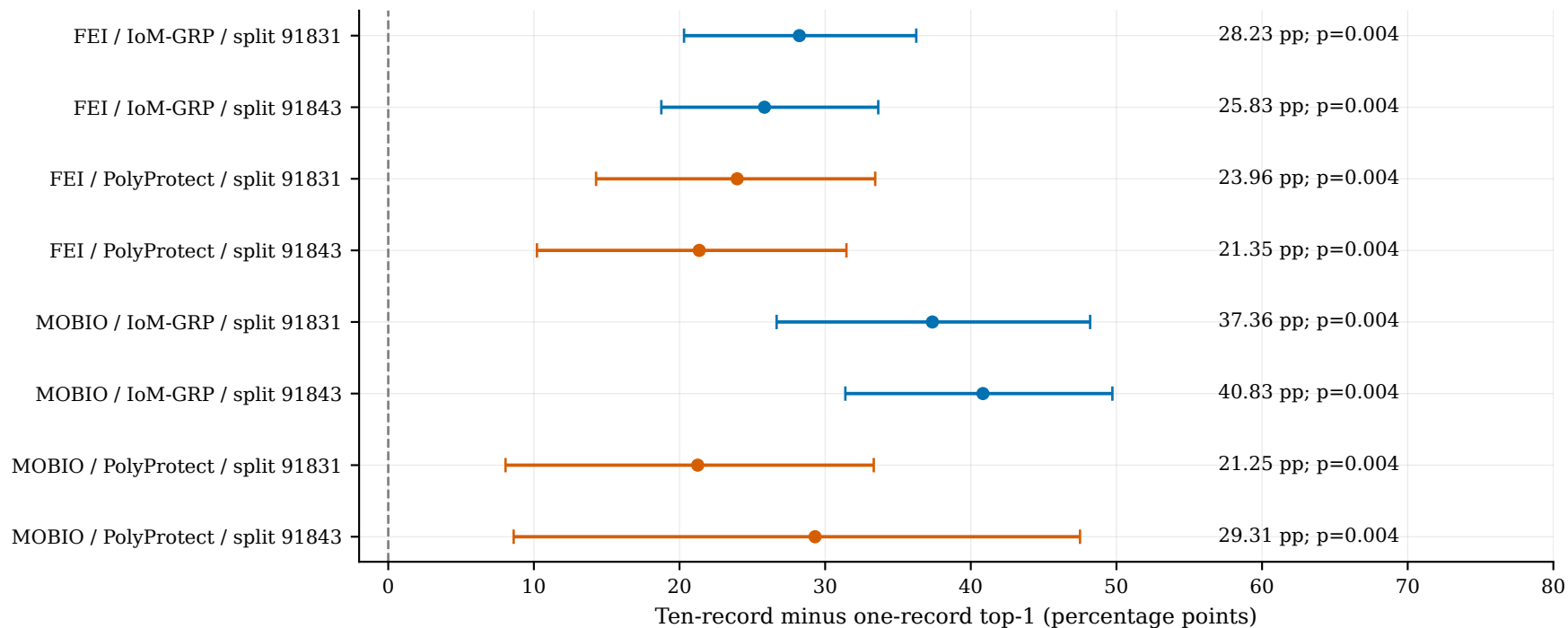


- 10 records, coarse sweep
- 10 records, fine sweep (new split)
- 1 record, coarse

(c) Fresh keys, image control



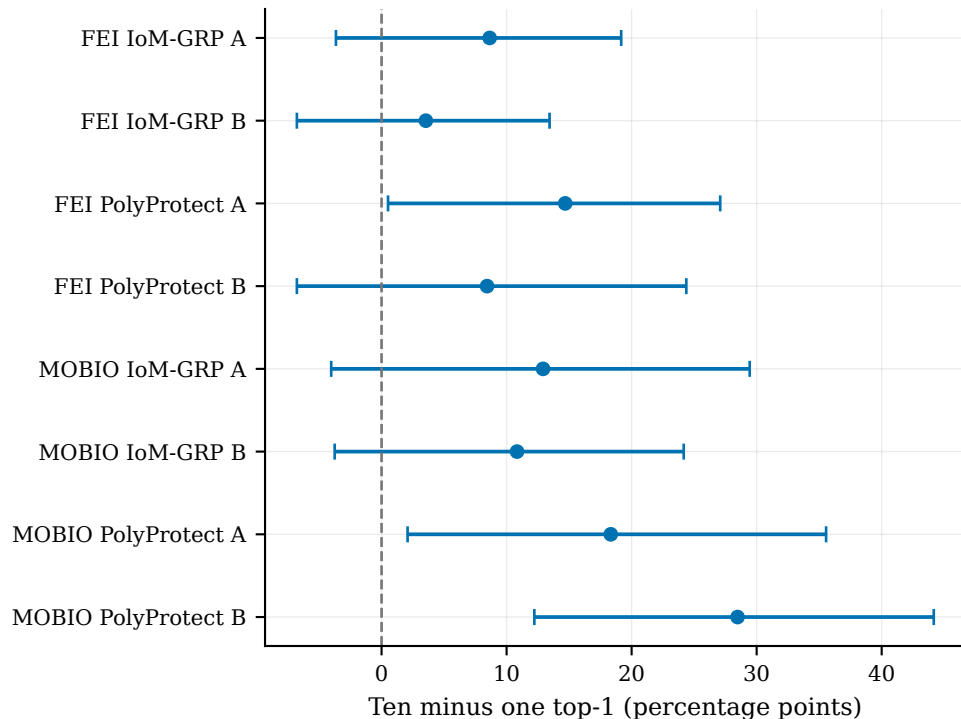
Pool-4 mean-pool amplification: three seeds, two identity partitions



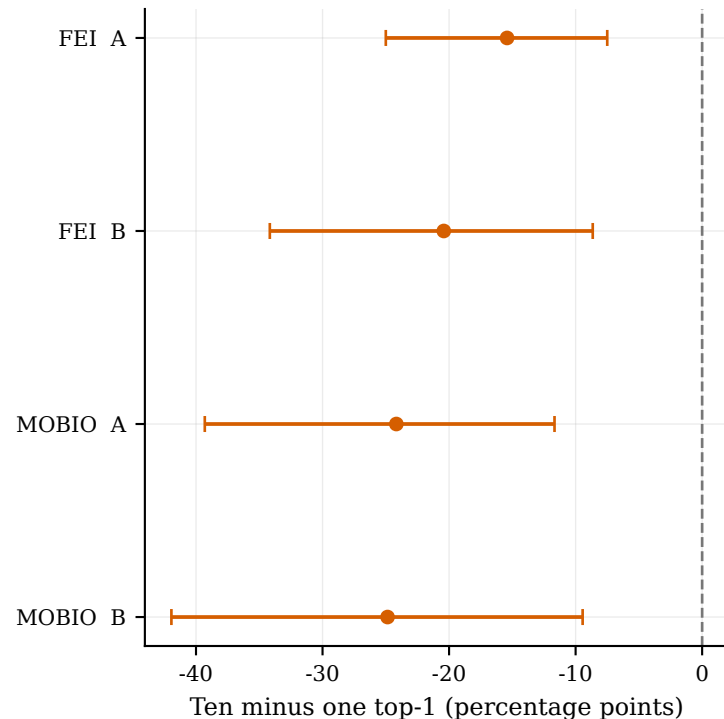
Bars: crossed model-seed / identity bootstrap 95% intervals. Tests: identity sign-flips, Holm family of 8.

Matched 120-epoch caps; key/set seeds fixed. Partitions are separate sensitivity analyses, not independent replications.

Pool-4 DeepSets: uncertain gains



Shared-key PolyProtect: regressions

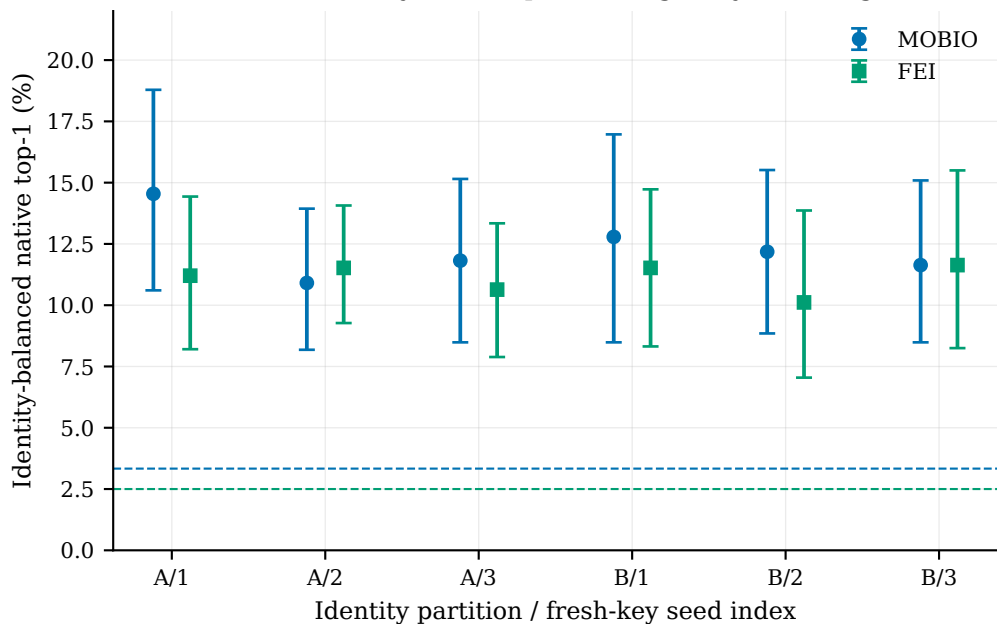


All 48 contrasts reported; post-hoc two-sided Holm family 48. Pointwise crossed seed/identity 95% intervals.

Left: none survives correction (minimum $p = 0.0576$). Right: all four negative contrasts survive ($p = 0.0192$).

All eight pool-4 mean-pool gains remain significant ($p = 0.0192$). More records do not universally improve linkage.

(a) Fresh PolyProtect: protected-gallery matching

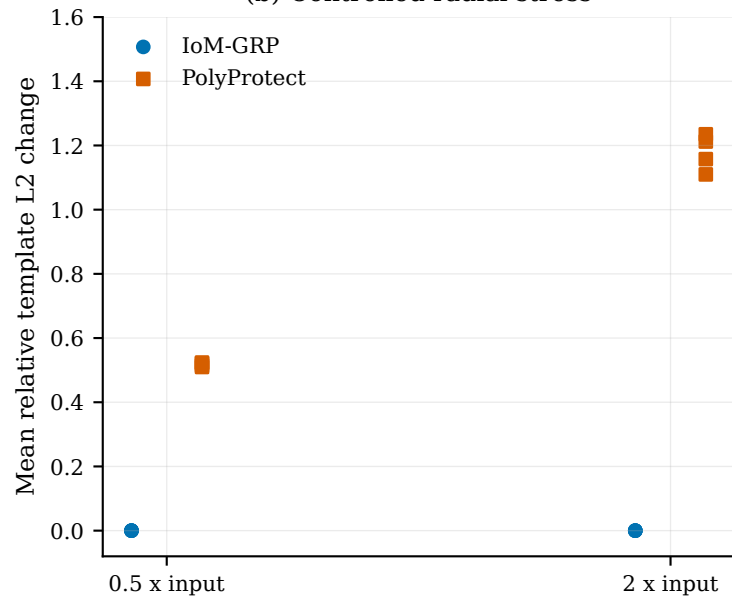


Native tests: 12 / 12 Holm-adjusted $p \leq 0.006$; identity-clustered 95% intervals; dashed lines: chance.

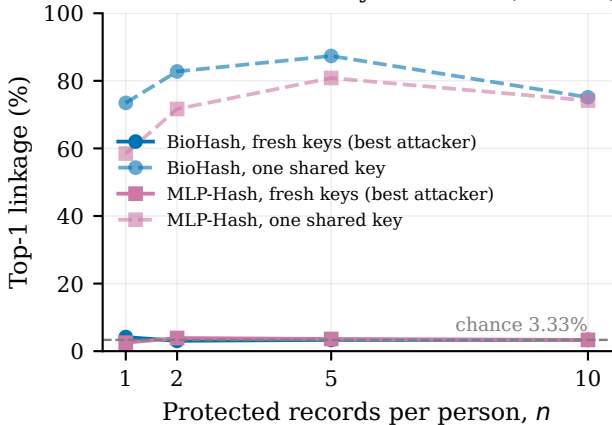
A/B: splits 91831/91843. Key seeds 1/2/3: 91873/91879/91883. Gallery-label permutations preserve probe clusters.

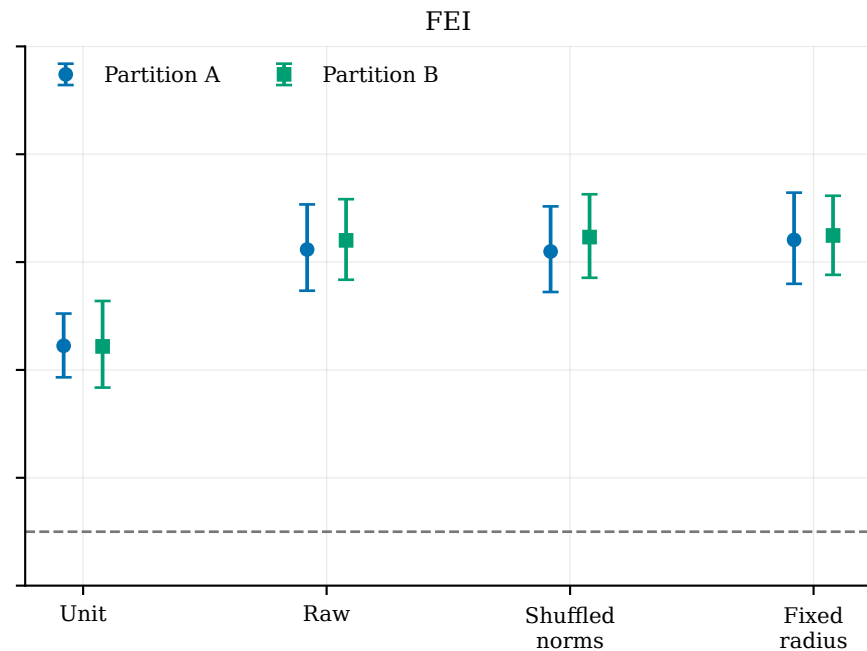
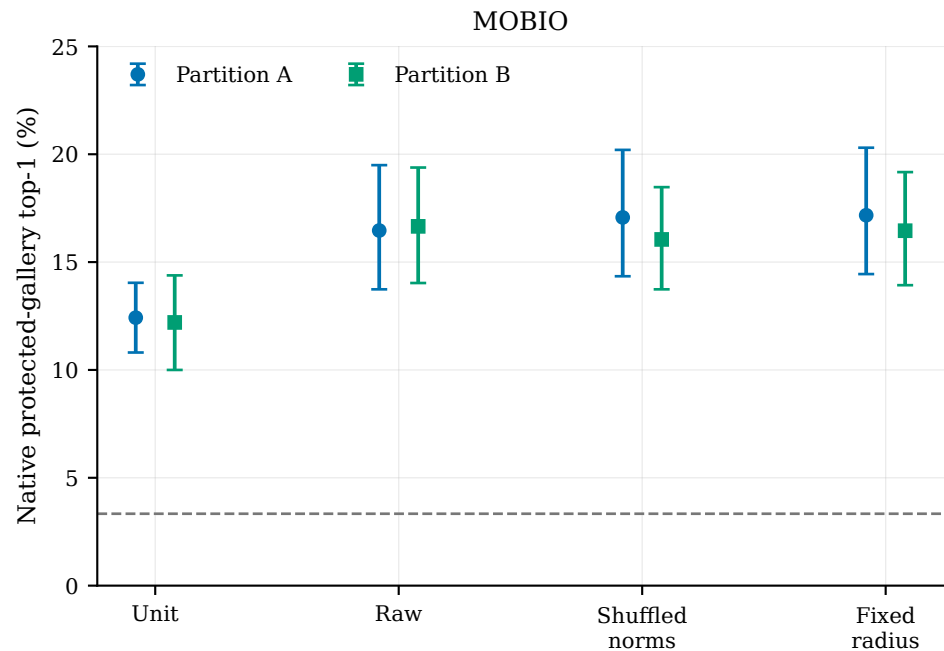
Radial stress: 32 held-out records per dataset/partition, fixed keys; measures scale sensitivity, not natural norm leakage.

(b) Controlled radial stress



Fresh and shared keys: MOBIO (N = 30)



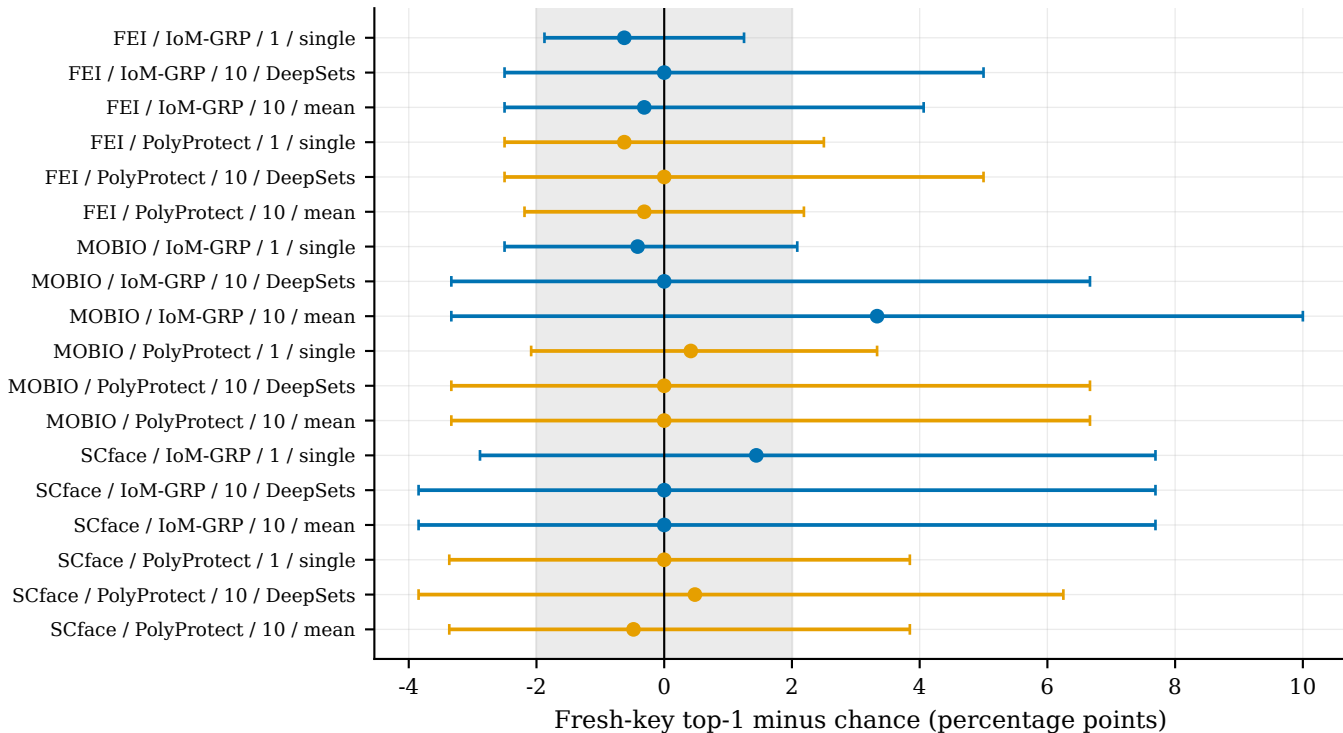


Three fixed key seeds averaged; identity-bootstrap 95% intervals. All 16 null tests: Holm $p = 0.0032$.

Raw minus unit survives paired correction only on FEI. Raw minus shuffled: all adjusted $p \geq 0.7584$.

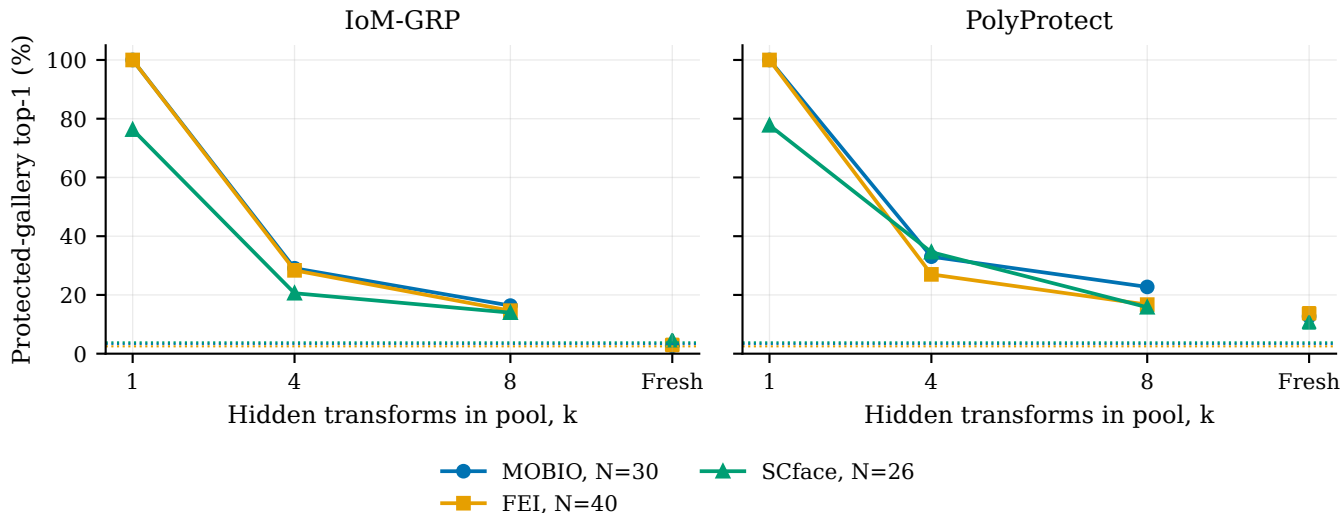
Natural identity-specific norm leakage is not established. No learned raw-input attack was retrained.

Fresh-key 90% intervals: exploratory pilot sensitivity

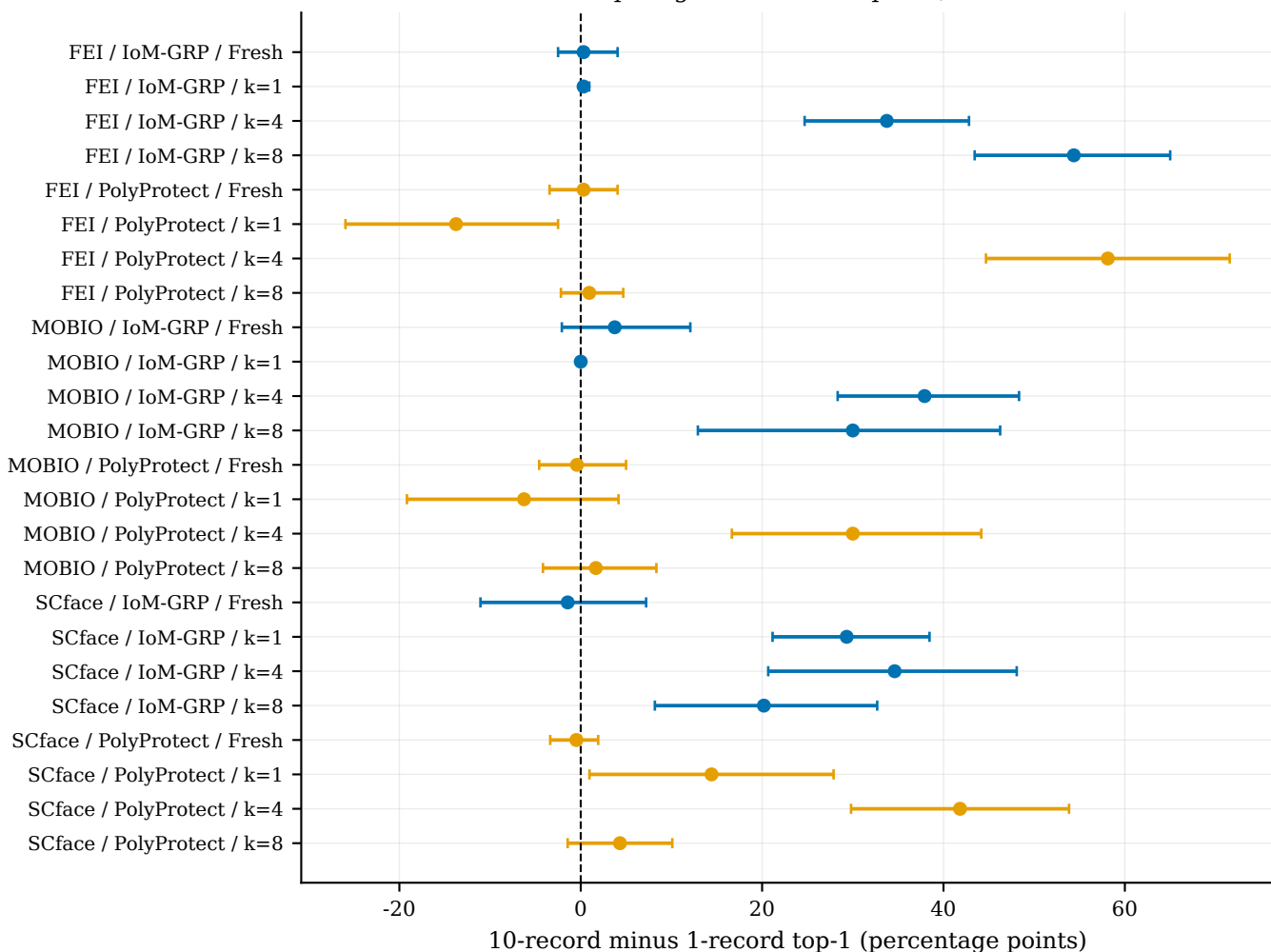


Illustrative +/-2 pp band; not an approved equivalence margin

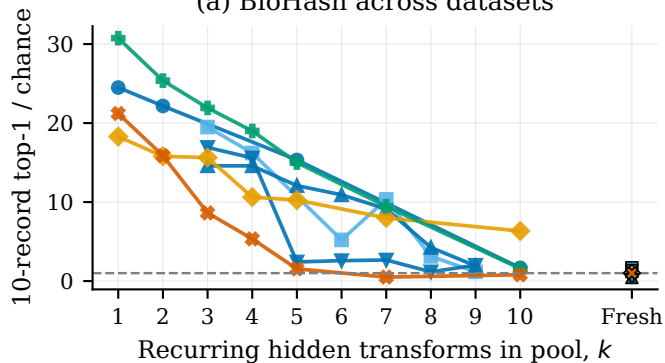
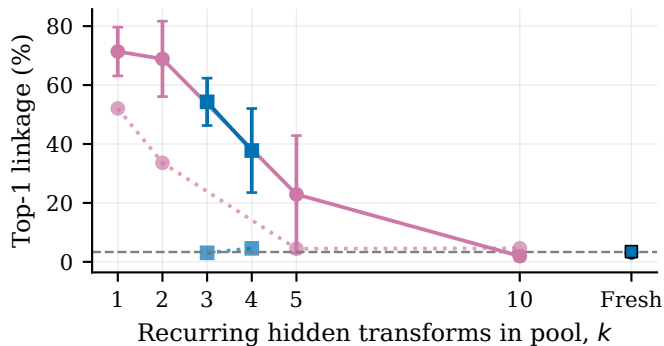
Native matching diagnostic, not learned embedding linkage



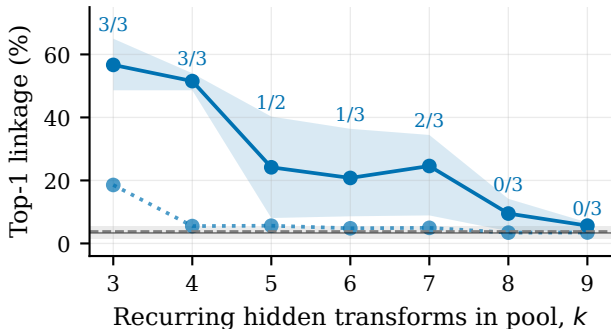
Paired mean-pool gains: one-seed pilots, 95% intervals



(a) BioHash across datasets

(b) MLP-Hash, MOBIO ($N = 30$)

MOBIO boundary (N = 30)

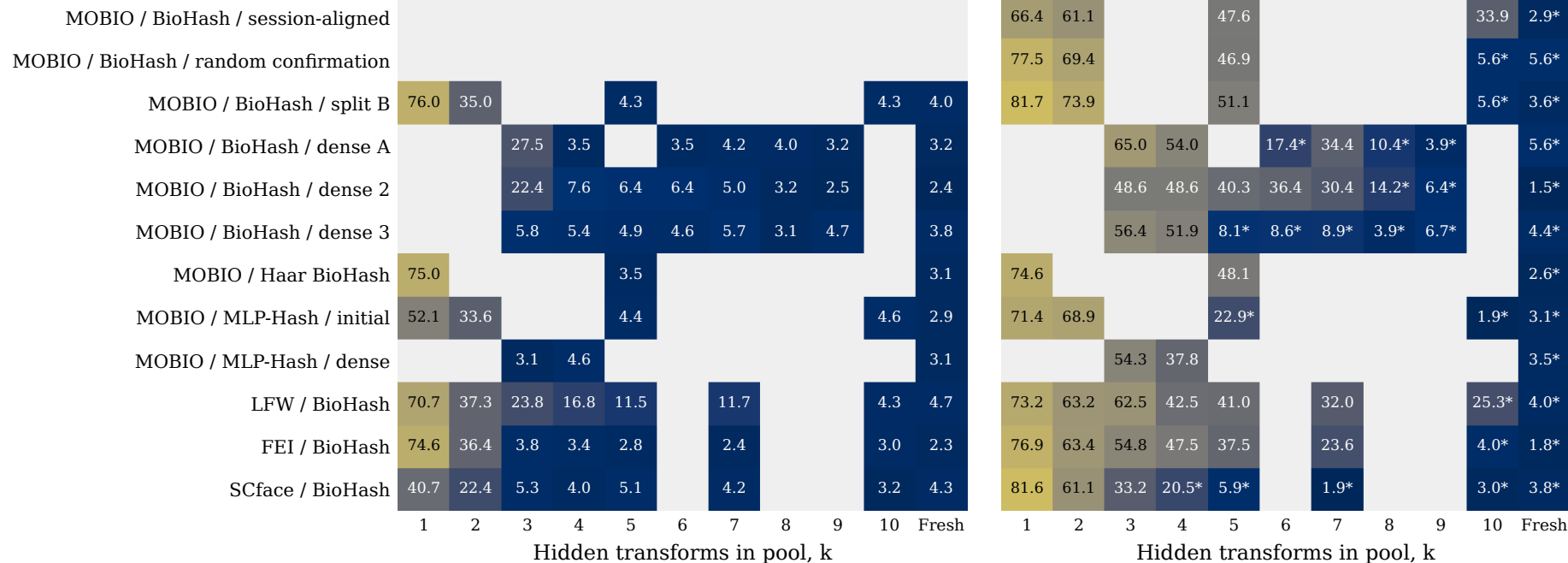


- range over available partitions
- 10 records, pooled mean
- 1 record, pooled mean
- fresh keys (3 partitions)
- chance 3.33%

Earlier three-seed key-pool studies

(a) One record

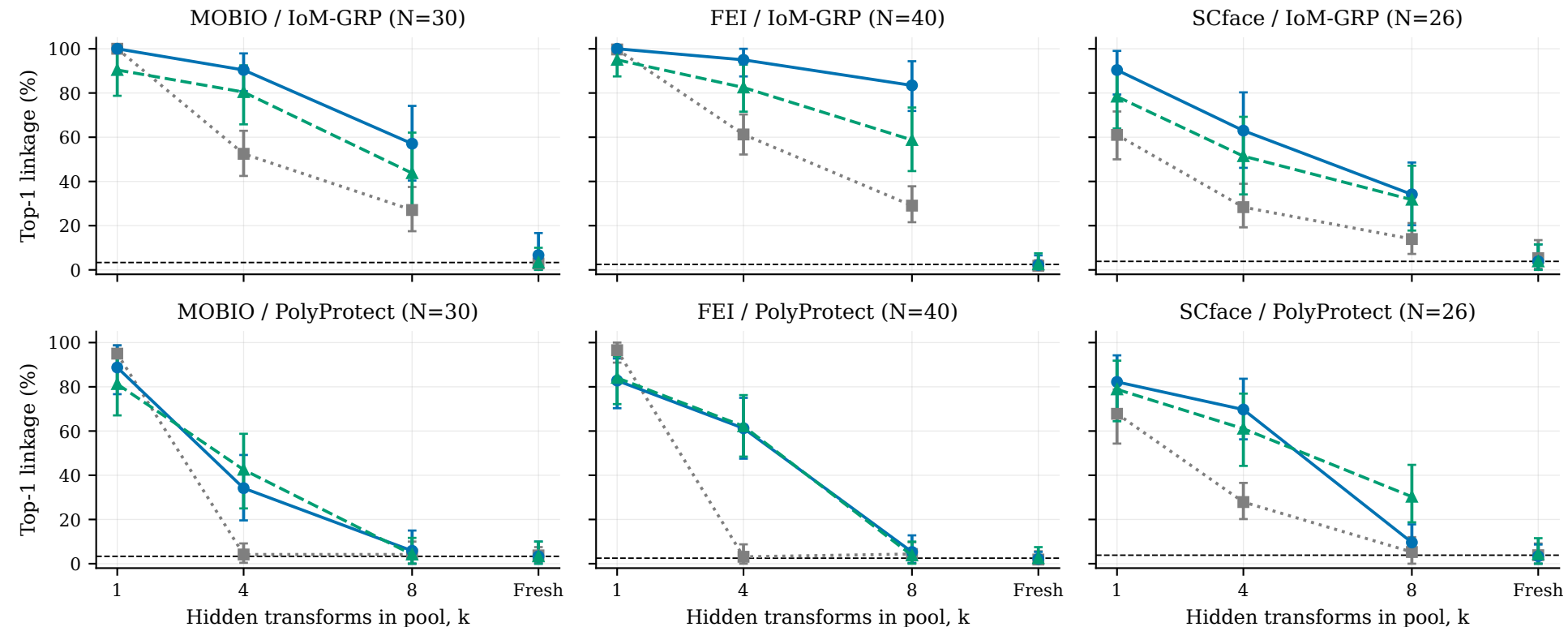
(b) Ten records



Cells: top-1 (%), 3-seed means. Grey: unavailable. *: not all seed intervals exclude chance.
 Gallery sizes: MOBIO 30; LFW 25; FEI 40; SCface 26. Rows are separate studies, not matched replications.

New protection schemes: one-seed pilots

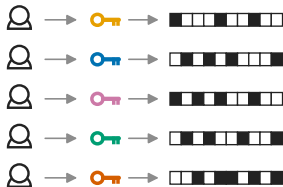
120-epoch cap; bars: identity-clustered 95% intervals; dashed baseline: chance



1 record 10 records, mean 10 records, DeepSets

Fresh keys

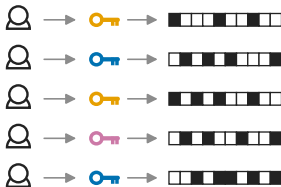
one key per source record



no transform reuse

Recurring key pool

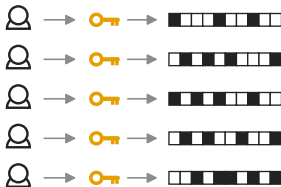
k transforms; slot labels hidden



same colour = same transform
reuse across train and test

Single shared key

k = 1; key value still hidden



shared-transform baseline

Learned attacks inconclusive

PolyProtect outside the theorem

Multi-record gains observed

boundary depends on dataset and split

Strong linkage observed

hidden does not mean unlearnable

Protocol

Disjoint train / validation / test identities; one held-out gallery image per identity
8 sets per identity; 3 seeds earlier, 1 in pilots; identity-clustered intervals